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Characterizing response quantity on academic social Q&A sites: a multidiscipline comparison of linguistic characteristics of questions

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Abstract

Purpose – As academic social Q&A networking websites become more popular, scholars are increasingly using them to meet their information needs by asking academic questions. However, compared with other types of social media, scholars are less active on these sites, resulting in a lower response quantity for some questions. This paper explores the factors that help explain how to ask questions that generate more responses and examines the impact of different disciplines on response quantity.

Design/methodology/approach – The study examines 1,968 questions in five disciplines on the academic social Q&A platform ResearchGate Q&A and explores how the linguistic characteristics of these questions affect the number of responses. It uses a range of methods to statistically analyze the relationship between these linguistic characteristics and the number of responses, and conducts comparisons between disciplines.

Findings – The findings indicate that some linguistic characteristics, such as sadness, positive emotion and second-person pronouns, have a positive effect on response quantity; conversely, a high level of function words and first-person pronouns has a negative effect. However, the impacts of these linguistic characteristics vary across disciplines.

Originality/value – This study provides support for academic social Q&A platforms to assist scholars in asking richer questions that are likely to generate more answers across disciplines, thereby promoting improved academic communication among scholars.

Keywords Response quantity, Linguistic characteristics, Academic questions, Academic social Q&A, Information seeking, Information communication

Paper type Research paper

1. Introduction

Social media platforms are frequently used to acquire knowledge, find solutions to problems and share opinions. Academic social networking sites (ASNSs) are a form of social media

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used by scholars (Van Noorden, 2014) that offer an online platform to support various activities in the research process, including allowing users to create personal archives with academic attributes, upload published papers and create online discussion groups. The more popular ASNSs include ResearchGate.net, Academia.edu, Mendeley.com and Zotero.org, among which ResearchGate.net is the largest (Van Noorden, 2014). The emergence of this type of online venue has provided scholars with an informal academic exchange platform, allowing them to overcome the limitations of geographical space, facilitating the exchange of academic information, and providing a location for storing academic information (Ovadia, 2014; Jeng *et al.*, 2017). The usual way to acquire academic information continues to be through reading published academic papers or books, and this new method needs time to develop. Therefore, a body of research on scholars' motivation for joining ASNSs is emerging that explores how to improve the popularity of these services (Muscanell and Utz, 2017). One of the services provided by ASNSs is a social Q&A function that allows communication among scholars on academic questions and answers. ResearchGate Q&A is the most representative and well-known academic social Q&A platform (Van Noorden, 2014). We found that the number of responses to questions on ResearchGate Q&A is smaller compared to other platforms (Jeng *et al.*, 2017), thereby hindering the prosperity and development of such ASNS services. To address this problem, we explored the characteristics of questions that affect response quantity.

One approach to exploring this topic is to study the factors affecting response quantity by looking at the linguistic characteristics of the questions. A previous study on social media found that "the phrasing of questions posed to one's social network directly influences the online responses received" (Teevan *et al.*, 2011). However, the linguistic characteristics affecting response numbers for questions on academic social Q&A networking sites have yet to be investigated. Moreover, previous research on academic social Q&A found that the characteristics of the question content differed significantly across disciplines (Jeng *et al.*, 2017). Therefore, this study compares the impacts of the linguistic characteristics of questions from different disciplines on their response numbers.

Based on the subject classification system devised by Abbott (2001), representative topics were selected under the five levels of hierarchy (humanities, social sciences, natural sciences, formal sciences and applied sciences). Question data sets were collected from ResearchGate Q&A, and an array of linguistic analysis methods was used to examine the words that scholars employ in their questions to determine which linguistic characteristics affect response numbers. Subsequently, multiple statistical analysis methods were used to establish the relationship between the linguistic characteristics of questions and response numbers, and to compare the differences across disciplines. This paper answers the following research questions:

RQ1. What linguistic characteristics of an academic question significantly influence its response quantity?

RQ2. How do the linguistic characteristics that influence the response quantity of questions vary across disciplines?

To the best of our knowledge, this is the first work to explore the factors affecting response numbers for academic questions from the perspective of their linguistic characteristics across disciplines. This study is intended to assist scholars in asking questions that attract more answers on academic social Q&A platforms.

2. Related work

2.1 Response numbers on social Q&A

A prerequisite to the success and continuous development of social media is that users continue to participate and contribute high-quality content (Jin *et al.*, 2015). The classic

legitimate peripheral participation of online community theory proposed by Lave and Wenger (1991) pointed out that the important first step for a prospective user to join the community is the community content they observe. For communities such as social Q&A, numerous studies have used empirical research to identify various types of intrinsic motivation (e.g. enjoying helping, knowledge self-efficacy, learning, and self-presentation) and extrinsic motivation (e.g. rewards in reputation system and peer recognition) that drive users to contribute answers (Jin *et al.*, 2013, 2015; Lou *et al.*, 2013). Another question that follows is how to evaluate high-quality answers. Although a wealth of research explores how to identify high-quality answers on social Q&A, questions are also a type of content in this space, and they are an important aspect that affects users' participation (Gazan, 2011). Furthermore, the basic premise of providing rich and high-quality answers should also include the quality of the questions (Teevan *et al.*, 2011). Therefore, the kinds of questions that attract more users to join the social Q&A community and answer questions is an important topic that needs to be studied, and the existing research does not explore this problem in depth.

At present, the research on this problem focuses on exploring the factors that affect the quality or success of the questions, and the number of responses to questions on social Q&A is used as a measure of the quality and success of the questions. For example, Yang *et al.* (2011) compared the differences between unanswered questions and overall questions considering the topic and heuristic features of questions (e.g. question length, history of the questioner and question posting time) on Yahoo! Answers. These features were used to predict whether the questions would be answered, but the F1-measure values of the predictions were low, showing that the predicted performance was not ideal. Choi *et al.* (2013) investigated questions on Yahoo! Answers using two classification criteria: "answered" and "not answered." Textual factors, such as whether a question starts with the word "what," the number of unique words and the presence of taboo words, were used to predict the success of questions. These were found to be significant factors in predicting whether or not the question would be answered. Another study (Chua and Banerjee, 2015) used content analysis to compare content features (e.g. level of detail, specificity and accuracy) of answered and unanswered questions on Stack Overflow; it found significantly different features between the two types of questions.

A small number of studies have explored the factors affecting the response quantity of Yahoo! Answers and Stack Overflow questions, finding that the factors affecting response quantity from the perspective of question content are the most important aspects considered in all studies. However, current research only puts forward a small number of relatively superficial content features of questions. There is no exploration of the problem from the perspective of language use style, which is a more in-depth representation of question content. Moreover, for academic questions, language use style may have more unique characteristics. In addition, these studies only distinguished response quantity according to whether the question was answered or not, and did not distinguish the level of the number of answers to the questions. In addition to the questions that have not received a response, social Q&A platforms are flooded with low-response questions, and these low-response questions need to be treated differently.

2.2 Academic social Q&A sites

ASNSs, as relatively new professional online social platforms for scholars, have many functions that are still being improved. Dimensions explored in research include the measurement of academic influence using altmetrics (Sugimoto *et al.*, 2017; Thelwall and Kousha, 2014, 2015), the understanding of the current status of ASNS use (e.g. use by people, use motivation, frequency of use, functions used, etc.; Jeng *et al.*, 2015; Muscanell and Utz, 2017;

Nández and Borrego, 2013), the factors affecting the use of ASNSs (e.g. disciplines, demographic characteristics, etc.; Jeng *et al.*, 2015; Ortega, 2015), and the academic social network structure on ASNSs (Jordan, 2014; Goodwin *et al.*, 2014). There is also a wealth of specialized research on aspects of the Q&A service platform provided on ResearchGate – ResearchGate Q&A – including the exploration of Q&A communication modes (Goodwin *et al.*, 2014; Jeng *et al.*, 2017) and the quality of academic answers (Li *et al.*, 2018, 2020c, d). However, most of the research on ASNSs mainly focuses on the exploration of the current situation, and there is less research on how to make ASNSs better serve academic information exchange among scholars.

Social Q&A is typical of online social references. How to improve the effectiveness of its services is the fundamental problem that needs to be solved for all social Q&A research (Gazan, 2011). Shachaf (2010) proposed an input–process–output (IPO) framework that abstracts the elements involved in information services of social Q&A in which the existence of communication norms is clearly pointed out in the input variable. The information behavior itself needs to be rooted in the norms of the community (Burnett *et al.*, 2001), and in addition to the codified norms, there are also norms that are generated in the daily information exchange process and that need to conform to the characteristics of the specific community (Burnett and Bonnici, 2003). Burnett *et al.* (2001) proposed the theory of normative behavior for communities; they noted that each community has its own unique environment and serves a specific group of people according to a set of established manners. It can be seen that each type of Q&A community needs its own code of conduct to ensure communication among community members, and the questioning manner is one of the most important aspects. As the initiator of the Q&A community input, the questioner abides by the corresponding rules and asks high-quality questions, which is the basis for ensuring the efficient progress of the entire communication process. As an academic, social and professional Q&A platform, ResearchGate Q&A is different from other social media platforms in terms of published content and users. Regarding content, academic questions are professional and may have wordings and descriptions that are different from generic questions. The users, questioners and responders are scholars with professional knowledge in a certain discipline (Li *et al.*, 2020c, d), and the way that information is expressed has its own discipline-specific requirements. Therefore, for this type of academic social Q&A community, it is necessary to explore the manner in which the questions are expressed.

A few related studies use the number of responses as the indicator to explore the manner of expression for academic questions. For example, Deng *et al.* (2019) coded the characteristics of questions on ResearchGate Q&A using the act4teams coding scheme proposed in the group interactions study; that is, they coded whether the content of the question is problem-focused, procedural, socio-emotional, or action-oriented, and used regression statistics to analyze the relationship between these characteristics and the number of answers. The results showed that the coded content characteristics of the questions raised from the perspective of information interaction were not related to response numbers. Because the content characteristics proposed in that study may not be significant factors affecting response quantity, further research is needed. Li *et al.* (2019) explored the response numbers for 130 questions on the Library and Information Services subject on ResearchGate Q&A, proposing characteristics from the perspective of the questioner and the question content. Regression analysis results showed that the characteristics of the questioner do not affect the response numbers for the question, but whether the question content provides resources and personal opinions does affect the question response numbers. However, this research only uses content analysis to analyze a small number of characteristics of a small amount of question content. Li *et al.* (2020b) used the linguistic characteristics of the question to predict the high or low response rates of a question on ResearchGate Q&A but found that the prediction accuracy was low and the prediction effect was not ideal. Understanding the

response quantity of a question is a complex matter, and there may be many other factors at work.

In research on general social media, it is noted that the topic of the question significantly affects the response rate of the question (Liu and Jansen, 2013). There have also been previous studies on academic social Q&A that have identified many differences in the expressions used across disciplines (Li *et al.*, 2020a). For example, Jeng *et al.* (2017) used content analysis to analyze the Q&A characteristics of the three disciplines in ResearchGate Q&A, and pointed out the differences in the expression of the content of the questions in different disciplines. For example, questions in the humanities express personal views and experiences more subjectively, and those in the natural sciences provide more factual information. Li *et al.* (2020c) pointed out that the evaluation criteria used by scholars are different when considering the quality of answers in different disciplines on ResearchGate Q&A. Using the Kruskal–Wallis test, Li *et al.* (2020a) confirmed that the linguistic features of the five subject questions on ResearchGate Q&A have significant differences in multiple language features (e.g. analytical thinking, function words, pronouns). Thus, different disciplines may have their preferred language expression styles, so the discipline may be an important factor affecting the response quantity for academic questions, and further research is needed.

In summary, there is some research on the response rates for academic questions. However, existing studies have not found more content characteristics that significantly affect the response rates for questions; many consider very little data from the perspective of a single subject, and others do not consider the subject to be an important factor in exploring this issue. Therefore, the present study makes up for the deficiencies of the above research, proposes more features at the level of question content from the perspective of language features, explores which features affect question response rates, divides question response rates into multiple categories, and distinguishes between disciplines to conduct in-depth research on this topic.

2.3 Linguistic characteristics

The simple analysis of the linguistic characteristics of text based on counting and categorizing words has been widely used to analyze the linguistic characteristics of text. It has been valuable for textual analysis in fields such as psychology, to measure social and psychological status (Tausczik and Pennebaker, 2010); pedagogy, to analyze changes in the linguistic characteristics of textbook content over time or to characterize students' self-reflections in their self-reported text (Gašević, 2014; Sell and Farreras, 2017); and marketing, to analyze customers' preferences and emotional states (Del Pilar Salas-Zárate *et al.*, 2015). With the continuous growth of social media, this method is also used in information science to analyze various types of user-generated content on social media, for example, to predict the personalities of users (Golbeck *et al.*, 2011) and differences in word usage based on the different personality types, ages and genders (Schwartz *et al.*, 2013). Different ways of thinking and the characteristics of different groups result in differences in language use styles, leading us to conclude that, on social media, academic questions using a particular language style are preferred by many scholars. Using statistical analysis to identify questions with this preferred language style could help scholars succeed in their academic communication.

Previous studies have also performed a simple linguistic characteristics analysis of questions on social Q&A to support related research. For example, Liu *et al.* (2011) used linguistic features to differentiate the types of questions on Yahoo! Answers to identify the information needs of users. Some studies of response quantity considered the content features that could affect response quantity by analyzing a small number of linguistic features of the question content. For example, Choi *et al.* (2013) found that whether questions

on Yahoo! Answers start with "what," the number of unique words, the level of clarity and the presence of taboo words were the most significant features for predicting whether or not a question would receive a reply.

Previous studies of linguistic characteristics for response quantity have only involved a minimal number of relatively simple linguistic characteristics, so there may well be other phrasing variants that are worthy of exploration. This study uses the Linguistic Inquiry and Word Count (LIWC), a tool proven in various research fields to provide accurate, objective, and rich linguistic characteristics to support research and to explore in depth which language use modes can generate more responses.

3. Methods

3.1 Data set

The data for this research were sourced from ResearchGate Q&A, a sub-platform provided by ResearchGate, the most influential ASNS in the world (Van Noorden, 2014). According to a large-scale survey of the journal *Nature* and multiple empirical research results, the disciplines of 20 million researchers from over 190 countries registered on ResearchGate include all disciplines, including the natural sciences, engineering, the humanities, social sciences and so on (Yan *et al.*, 2021; Van Noorden, 2014; researchgate.net/about). Due to the differences in the number of registrations, activity levels and participation motivations of scholars from different disciplines on ResearchGate, the data used in the published research show that the monthly average number of questions asked under different research topics on ResearchGate Q&A ranges from 0.8 to 1.5 (Jeng *et al.*, 2017). Thus, scholars do not ask many questions, and they are more willing to participate in answering than asking questions (Huang *et al.*, 2019). This can also illustrate the importance of this research.

To ensure the representativeness of the data and to compare differences in the linguistic characteristics of the questions that affect response quantity across disciplines, we selected several disciplines as research objects from ResearchGate Q&A based on the following two criteria. First, we chose disciplines that had questions with many answers, as well as questions with few answers. Second, the selected disciplines had a similar number of questions. The five disciplines we chose are linguistics, applied psychology, mathematical physics, data science and public health. These disciplines represent, respectively, branches of the humanities, social sciences, natural sciences, formal sciences and applied sciences. In addition, we had established from previous studies that the questions in these five disciplines differ significantly in terms of many linguistic characteristics (Li *et al.*, 2020a), offering the possibility for further exploration of our research topic.

ResearchGate Q&A uses tags marked by the questioner as the basis for classifying the discipline to which the question belongs, and the website has a homepage for each discipline. However, we found that each discipline homepage only provides a limited number of recent questions or recently answered questions. Therefore, we used the search function provided by ResearchGate to find questions containing tags for each discipline, and we collected the URLs. When 20 consecutive questions without a discipline tag were retrieved, we regarded the match between the search results and the retrieval requirements to be low – meaning that we were unable to obtain further target questions – so we stopped collecting URLs. We used the above method to collect question data for each discipline, eventually collecting 448 questions under linguistics, 339 under applied psychology, 375 under mathematical physics, 328 under data science and 478 under public health. A total of 1,968 question URLs were collected. Simultaneously, on August 26, 2019, we used a crawler program to acquire the question title, content, number of answers and question posted time for the question URLs within each discipline.

ResearchGate Q&A does not provide an API to collect all question data for each discipline. Therefore, it was not possible to collect all of the available questions. However, the method

used was similar to random sampling and ensures the representativeness of the data. The questions on each topic were sorted according to the posted time and content relevance, with no priority given to questions with many responses or having received a high degree of attention. This data collection method has also been applied to other related studies that require question data on ResearchGate Q&A (Jeng *et al.*, 2017; Deng *et al.*, 2019). As shown in Table 1, we also calculated the average number of responses to questions for each discipline and all questions we collected, similar to what is reported in other studies (Jeng *et al.*, 2017; Deng *et al.*, 2019).

3.2 Variable acquisition

3.2.1 Acquisition of characteristics. First, we combined the title and content of each question as an object for analyzing the linguistic characteristics before using the latest version of the LIWC (LIWC2015) as the linguistic analysis tool to extract these characteristics. The LIWC uses multiple dictionaries to classify words and to analyze the linguistic characteristics (Pennebaker *et al.*, 2015; Tausczik and Pennebaker, 2010). LIWC2015 has a multi-step dictionary construction method, which ensures that the constructed dictionary is more internally consistent than in the previous version (Pennebaker *et al.*, 2015). The question text was imported into LIWC2015; the program matched each word in each question with the predefined dictionary categories for multiple linguistic characteristics and calculated the percentage of words in each characteristic linguistic category for each question. LIWC2015 also considers various structural composition elements (e.g. word count and sentence punctuation) during word matching. The software can gather 93 linguistic text characteristics, including four summary language variables (analytical thinking, clout, authenticity and emotional tone), four general descriptor categories (word count, words per sentence, percentage of target words captured by the dictionary and percentage of words in the text that are longer than six letters), 21 standard linguistic dimensions (e.g. percentage of words in the text that are pronouns, articles, auxiliary verbs, etc.), 40 word categories tapping psychological constructs (e.g. affect, cognition, biological processes, drives), six personal concern categories (e.g. work, home, leisure activities), six informal language markers (e.g. assents, fillers, swear words, netspeak) and 12 punctuation categories (e.g. periods, commas, etc.; Pennebaker *et al.*, 2015).

3.2.2 Response quantity confirmation. Response quantity was measured as the number of answers to a question. As existing research has found that it takes longer to obtain the first response to a question on ResearchGate Q&A than on other types of social media (Jeng *et al.*, 2017), responses to academic questions can take a long time to accumulate. Therefore, we decided that using existing ways to determine response quantity on other types of social media – that is, the number of responses obtained within three days (Teevan *et al.*, 2011) or one week (Liu and Jansen, 2013, 2018) after the question is posted – are not suitable for academic questions. Instead, this study distinguished response quantity by calculating the average number of responses obtained in the month between the time of data collection and

	Response quantity			
Disciplines	Mean	Median	Std. deviation	*N*
Linguistics	10.19	6.00	21.657	448
Applied psychology	16.13	5.00	121.973	339
Mathematical physics	20.35	7.00	48.173	375
Data science	6.29	4.00	9.754	328
Public health	10.51	6.00	14.774	478
Total	12.58	6.00	56.530	1,968

Table 1. Descriptive statistics for the number of responses to questions in each discipline and all questions

the time the question was posted. The few questions that had been posted for less than one month were deleted. In this way, we ensured that the number of responses calculated was within the same timeframe, thereby avoiding any impact of the length of exposure time on the number of responses. Questions posted for more than one month with no responses were also deleted. Figure 1 shows the process of confirming the response quantity and the consequent changes in the data.

3.3 Data analysis

To not affect the efficacy of the analysis, we adopted the numerical data binning method used in feature engineering to partition the response quantity data. To avoid an effect of the difference in the overall response quantity across disciplines (due to differences in the number of scholars engaging on ResearchGate), the question data of the five disciplines were separated and independently determined the level of the number of question responses.

The distribution of response quantity for each discipline is shown in Figure 2a–e. All disciplines have a strong right skewness and do not conform to the normal distribution (the p-value of the Kolmogorov–Smirnov test is 0). We used the adaptive binning method to determine the range of the intervals according to the data distribution. Therefore, as Figure 2 shows, the quartiles divide the response quantity of each discipline into four intervals with an equal number of questions, and assigns values from 1 to 4, where 1 represents the lowest response quantity and 4 the highest. In the final data set, 464 questions have a response level of 1, 467 questions have a response level of 2, 463 questions have a response level of 3, and 464 questions have a response level of 4.

Logistic regression analysis was used to identify the linguistic characteristics of questions affecting response quantity. In this study, 93 linguistic characteristics obtained using the LIWC software were regarded as independent variables. These were processed by log transformation to ensure that they had the same dimensions. The four levels of response quantity were regarded as the dependent variables. A collinearity diagnostic test on the independent variables showed that either the tolerance of some independent variables was less than 0.1 or the variance inflation factor was greater than 10, indicating some collinearity. As the aim was to detect the relationship between all independent variables and the response quantity for the question, we did not delete collinear independent variables.

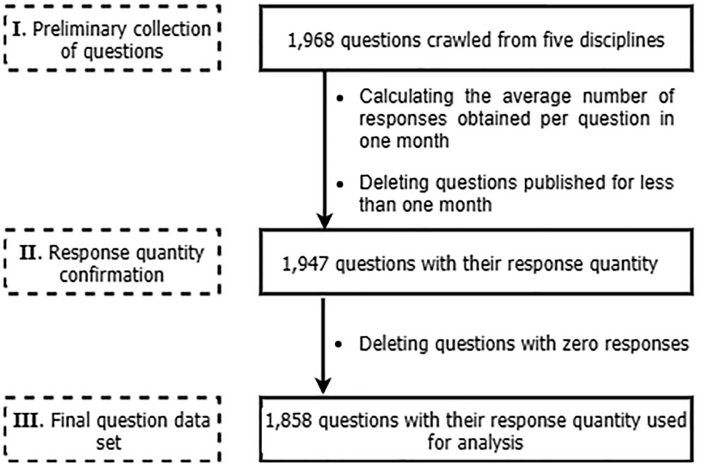
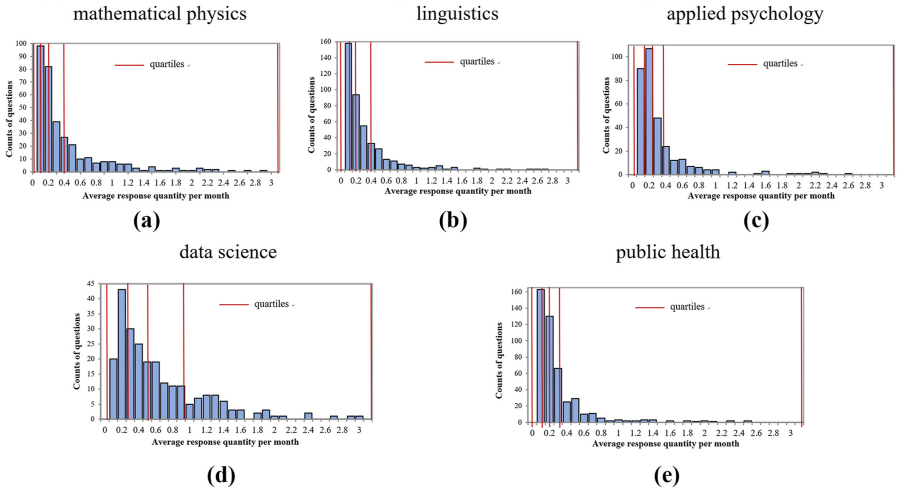


Figure 1.
Filtering the question data set for response quantity confirmation



Note(s): X-axis intervals of the histogram (a)–(e) are all 0.1

Figure 2.
Histogram of the response quantity for each discipline's questions with quantiles (frequency concentration part)

According to the relationship between correlation and regression analysis (i.e. that correlation analysis is the basis and premise of regression analysis), we used both methods to identify the linguistic characteristics of questions with a significant linear influence on response quantity. First, correlation analysis was used to identify independent variables that were significantly correlated with response quantity to eliminate irrelevant linguistic characteristics. Then, logistic regression was used to identify the independent variables with a significant linear effect on the dependent variable. Spearman's correlation was used because the dependent variable in this study was a categorical variable. Likewise, ordinal logistic regression (OLR) was used, as the dependent variables were also ordinal variables with multiple categories.

To compare the differences in the linguistic characteristics of questions affecting response quantity across disciplines, the same data analysis was used to analyze the data for each discipline separately. Comparing the data analysis results of different disciplines shows the differences between the five disciplines in how linguistic characteristics affect response quantity.

4. Results

4.1 Question linguistic characteristics and response quantity

The results of the correlation analysis in Table 2 show the linguistic characteristics of questions that are significantly associated with response quantity. Thirty-seven linguistic characteristics, sorted according to the absolute value of the coefficient, are significantly related to response quantity. The highest degrees of correlation were found for word count, dictionary words, future focus, second-person pronouns and commas. We performed a collinearity diagnosis on these linguistic characteristics. Table 2 shows that these characteristics do not exhibit multicollinearity and can be used for further OLR analysis. Table 2 shows that for this OLR model, the chi-square value of the parallel test is 66.313 and the p-value is 0.726, indicating that the model passes the parallel test, and the analysis result is credible. The model fitting information is χ^2 (37) = 160.502, p = 0.000, indicating that the model is effective. Table 2 highlights eight characteristics that have a significant linear correlation with response quantity.

Question linguistic characteristics significantly related to the response quantity		Correlation analysis		Collinearity statistics		Ordinal logistic regression	
		Coef.	p	Tolerance	VIF	Estimate	p
General descriptor categories	Word count	0.154**	0.000	0.475	2.104	0.677	0.000
	Dictionary words	0.148**	0.000	0.318	3.148	4.960	0.000
Summary language variables	Authenticity	0.050*	0.032	0.421	2.376	0.054	0.631
	Emotional tone	0.061**	0.009	0.173	5.780	–0.361	0.087
Standard linguistic dimensions	Total function words	0.085**	0.000	0.322	3.104	–2.143	0.005
	1st person singular	–0.055*	0.017	0.654	1.529	–0.509	0.000
	1st person plural	0.070**	0.003	0.651	1.536	0.207	0.395
	2nd person	0.098**	0.000	0.776	1.289	0.386	0.037
	3rd person plural	0.051*	0.029	0.843	1.186	0.086	0.736
	Articles	0.067**	0.004	0.708	1.412	0.252	0.116
	Auxiliary verbs	0.067**	0.004	0.510	1.962	0.458	0.102
	Negations	0.085**	0.000	0.725	1.380	0.217	0.344
	Common verbs	0.083**	0.000	0.496	2.016	0.255	0.390
	Comparisons	0.061**	0.009	0.801	1.249	–0.019	0.895
	Quantifiers	0.051*	0.027	0.770	1.299	0.007	0.961
	Positive emotion	0.092**	0.000	0.282	3.543	0.580	0.013
	Negative emotion	0.060**	0.009	0.238	4.205	–0.638	0.050
	Sadness	0.096**	0.000	0.796	1.256	0.896	0.027
	Social processes	0.047*	0.045	0.531	1.883	–0.249	0.118
	Friends	0.053*	0.022	0.866	1.155	0.313	0.381
	Tentative	0.049*	0.035	0.690	1.450	0.045	0.762
	Differentiation	0.080**	0.001	0.541	1.849	0.091	0.586
	Biological processes	0.065**	0.005	0.161	6.193	0.056	0.843
	Health	0.046*	0.048	0.163	6.138	–0.068	0.820
	Affiliation	0.055*	0.018	0.546	1.831	0.012	0.944
	Reward	0.048*	0.040	0.828	1.208	–0.187	0.267
	Risk	0.069**	0.003	0.721	1.388	0.310	0.232
	Past focus	0.088**	0.000	0.819	1.221	0.129	0.427
	Future focus	0.098**	0.000	0.912	1.097	0.218	0.264
	Relativity	0.070**	0.003	0.429	2.330	0.058	0.790
	Motion	0.097**	0.000	0.819	1.222	0.244	0.156
	Time	0.082**	0.000	0.683	1.465	–0.017	0.910
Informal language markers	Swear	0.046*	0.047	0.981	1.019	3.570	0.085
Punctuation	Periods	0.057*	0.015	0.798	1.253	0.126	0.405
	Commas	0.098**	0.000	0.789	1.268	0.259	0.049
	Colons	0.054*	0.020	0.874	1.145	–0.033	0.910
	Exclamation	0.050*	0.032	0.935	1.070	0.285	0.442

Note(s): ***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Table 2.
The results of correlation analysis, collinearity diagnosis and OLR analysis

In descending order of regression estimates, linguistic characteristics with a significant, positive, linear relationship to response quantity include dictionary words, sadness, word count, positive emotion, second-person pronouns and commas. The words of the question identified by the LIWC dictionaries have a positive impact on response quantity. This shows that, in this study, the dictionaries defined in LIWC for identifying linguistic characteristics are suitable to explain the relationship between these characteristics of questions and their response quantity. Moreover, the more sadness and positive emotions in the questions, the more responses will be obtained. Furthermore, question length does not reduce response quantity, and if a question contains more second-person pronouns or more commas, the response quantity will likely be higher.

In descending order of the absolute value of the regression estimates, the linguistic characteristics that were negatively and significantly correlated with response quantity

include the number of function words and first-person pronouns, indicating that there may not be many function words and first-person pronouns in high-response questions.

4.2 Question linguistic characteristics and response quantity across disciplines

Table 3 shows the linguistic characteristics of the questions in different disciplines that were significantly related to response quantity after performing the correlation analysis and collinearity tests for each discipline's data set. Estimates and associated *p*-values are listed separately for these significant relationships. Table 3 shows that the differences between the five disciplines are considerable. There is no significant relationship between a specific linguistic characteristic and the response quantity in the five disciplines. Question response quantities in up to three disciplines were significantly related to the same linguistic characteristics.

The general descriptor categories show that question length has a positive and significant impact on response quantity in public health, mathematical physics and applied psychology. In these three disciplines, the longer the question, the more responses received. In the linguistics and mathematical physics disciplines, questions including more words in the LIWC dictionary had a positive impact on response quantity. Questions in these disciplines attract more responses when they include words related to psychological concepts. In this category, for questions on mathematical physics, including more long words does not reduce response quantity.

In the standard linguistic dimensions category in the linguistic and mathematical physics discipline, including more first-person singular pronouns in the question negatively affects response quantity. In data science, second-person pronouns have a positive impact on response quantity.

In the psychological constructs category, data science only shows significant results for questions with biological processes; applied psychology only shows significant findings for questions with see-related words; in linguistics, questions expressing anger and positive emotions as well as risk descriptions have a higher number of responses; in public health, questions expressing positive emotions receive more responses; in mathematical physics,

Question linguistic characteristics		Data science		Applied psychology		Linguistics		Public health		Mathematical physics	
		Estimate	p	Estimate	p	Estimate	p	Estimate	p	Estimate	p
General descriptor categories	Word count	ns	ns	1.023	0.02*	ns	ns	1.000	0.022*	0.0934	0.047*
	Words > 6 letters	ns	ns	ns	ns	ns	ns	ns	ns	2.073	0.013*
	Dictionary words	ns	ns	ns	ns	3.624	0.006**	ns	ns	4.501	0.034*
Standard linguistic dimensions	1st person singular	ns	ns	ns	ns	−0.594	0.003**	ns	ns	−1.562	0.000**
	2nd person	1.055	0.018*	ns	ns	ns	ns	ns	ns	ns	ns
Psychological constructs	Positive emotion	ns	ns	ns	ns	0.388	0.031*	0.796	0.019*	ns	ns
	Anger	ns	ns	ns	ns	1.306	0.017*	ns	ns	ns	ns
	See	ns	ns	−1.276	0.006**	ns	ns	ns	ns	ns	ns
	Biological processes	1.236	0.032*	ns	ns	ns	ns	ns	ns	ns	ns
	Risk	ns	ns	ns	ns	1.334	0.034*	ns	ns	ns	ns
Personal concern categories	Leisure	−2.447	0.002**	ns	ns	ns	ns	ns	ns	ns	ns
Punctuation	Dashes	−0.902	0.02*	ns	ns	ns	ns	ns	ns	ns	ns

Note(s): ns = not significant; red = positive and significant; green = negative and significant; *$p < 0.05$, **$p < 0.01$

Table 3.
OLR analysis results of linguistic characteristics significantly related to response quantity across disciplines

there is no significant linear correlation between the number of responses to questions and linguistic characteristics. Finally, in the personal concern and punctuation categories, data science questions mentioning leisure and including dashes are negatively affected.

5. Discussion

5.1 Insights based on the findings

First, among the linguistic characteristics under the general descriptor category, word count has a different effect on the response quantity of academic questions and generic questions. For example, a study of Stack Overflow found that shorter questions were more likely to be answered (Chua and Banerjee, 2015). Teevan *et al.* (2011) also found that one-sentence questions on Facebook received more answers than two-sentence questions. This contrasts with the positive correlation found in this study between the word count of a question and its response quantity. Academic questions may be more difficult to understand, so a more detailed, longer description is needed for responders to understand an answer. Scholars may tend not to answer short or unclear questions. Furthermore, we found that, in the domains of mathematical physics, public health and applied psychology, the longer the question, the higher the response rate. The possible reason is that, compared to the other two disciplines, for clarity, the questions in these three disciplines need more content. Mathematical physics may require detailed descriptions of theories, and public health and applied psychology may require detailed descriptions of examples.

Second, among the linguistic characteristics under the standard linguistic dimension category, the characteristic of emotion affects the response quantity of academic questions and generic questions in the same way. Liu and Jansen's (2013) study of response quantity for questions on Sina Weibo found that language involving emotions had a positive and significant effect on response quantity. This is consistent with the findings of this study. Both positive and negative references to emotion in academic questions generated more responses, showing that scholars are more willing to answer questions expressing emotions. Furthermore, in public health and linguistics, we found that questions with emotional expressions receive more responses. Research analyzing health questions on general social Q&A platforms to examine health-information-seeking behaviors indicates that health questions often involve emotional experiences (Oh *et al.*, 2013; Bowler *et al.*, 2012). We also identified public health and linguistics questions in the data set with outstanding performance on this characteristic. Public health questions with an emotional narrative tend to elicit greater levels of concern and, thus, a higher number of responses. In addition, under this linguistic characteristics category, some linguistic characteristics that were not covered in studies of generic questions also affect response quantity; these characteristics may help to improve response quantity for academic questions. Specifically, we found that having too many function words and first-person pronouns in a question is not conducive to high response rates. However, second-person pronouns generate more responses. Personal pronouns are the most-used category of function words. Research has pointed out that using too many first-person pronouns can make a question too subjective (Shah *et al.*, 2014), which may affect the response rate. In academic questions, first-person pronouns usually should not express the author's situation and subjective thoughts; instead, if the question involves inviting the opinions of others, there will likely be more responses. Meanwhile, we also found that for disciplines in the humanities and those belonging to the natural sciences, using more first-person pronouns is not conducive to attracting responses. Moreover, in disciplines belonging to the formal sciences, using more second-person pronouns in questions increases the response rate. However, the reason for the effects of differences in pronoun use could be that scholars in some disciplines are indifferent to pronoun use, or because the use of some pronouns is not prominent in the data set, but this has not yet been determined, and further exploration is needed.

Next, in terms of punctuation, the use of more commas indicates that the sentence would be easier to read if it were shorter and more clearly expressed (Robinson *et al.*, 2013). This is an important point for academic questions. The use of short sentences is more conducive to understanding and is beneficial for response rates.

We also found many differences in other influencing factors. However, the lack of research on the response rate for academic questions across disciplines made it difficult to identify the reasons for these differences. We will need to further verify and explain these differences by surveying scholars or by expanding the data set.

5.2 Theoretical implications

There have been many studies on general social Q&A platforms, such as Yahoo! Answers, and fewer research studies on domain-based social Q&A platforms. With the gradual increase in social Q&A platforms in various domains, research in this area should be encouraged. At present, research on domain-based social Q&A platforms often focuses on health Q&A platforms, and research on academic social Q&A platforms has only recently appeared. Also, most of the previous studies on social Q&A platforms focus on evaluating the quality of answers, while ignoring, to a certain extent, differences in answer quality that might be caused by the questions raised. Therefore, this research studied an important aspect affecting the flourishing and development of academic social Q&A platforms, namely, the response quantity of the questions on these platforms.

As shown in Figure 3, for social Q&A, the most important information activity is the information exchange process of questioning and answering. This process needs to be carried out under the norms of social Q&A. The asker who asks the question needs to follow the norms of how to ask, and as the initiator of the information exchange process, the research of the norm is more important for the asker. This research explores the linguistic characteristics of academic questions by using the indicator of response quantity, thereby improving the relevant theories about norms of online communities from the perspective of questioning.

5.3 Practical implications

The question characteristics we found to affect response quantity could improve academic social Q&A platforms in the following practical ways. First, scholars find some questions to be complex and difficult to understand. The answers to such questions cannot be easily found in standard academic sources, and it can be time-consuming to locate information using search engines or by consulting the literature. From the perspective of the linguistic characteristics of a question, this study can guide questioners on how to ask better questions on academic social Q&A sites that will generate a large number of targeted professional responses, thereby saving time and effort and meeting needs for academic information.

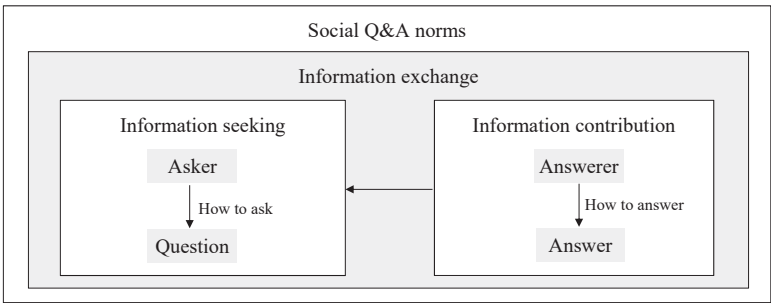


Figure 3.
Information exchange process of social Q&A

Simultaneously, more responses to questions could provide a wealth of information for other users of academic social Q&A platforms with the same information needs.

Second, on academic social Q&A platforms, providing a help center page or FAQ could assist questioners in attracting a high number of responses. Scholars could gain maximum benefits from academic social Q&A platforms, thereby increasing their willingness to join these platforms, ultimately increasing the prosperity of academic social Q&A networking sites. Specifically, the characteristics of questions that affect response quantity identified in this research could empower platform help centers to provide users with effective questioning skills. At the early stage of question posting, the platform should monitor questions that are likely to receive fewer responses and implement measures such as increasing the exposure time of the question or suggesting that the questioner modify the question.

5.4 Study limitations and future work

This study has several limitations that also represent areas that need to be improved and further explored. First, we should note that the only study of question responses on academic social Q&A platforms found a significant negative correlation between the length of the question and the number of responses (Deng *et al.*, 2019), contradicting the findings of this study. As there are few other related studies, our findings need to be verified by expanding the data set.

Second, techniques that only involve matching words are relatively crude and cannot capture the full meaning of questions. However, this paper highlights aspects for follow-up research to carry out more verification. Therefore, future research could be based on the significant characteristics found in this study. Aspects needing further content analysis could be included to verify the findings presented here.

Third, additional in-depth research is needed to compare the differences across disciplines. Although this study obtained data sets from five disciplines, its findings are still not comprehensive, and the reasons for some differences need further exploration. Therefore, in future research, other research methods, such as scholar surveys, content analysis or expansion of the domain data set, should be explored.

Fourth, this research focused on the ResearchGate Q&A platform and did not consider academic questions on other types of social media. Therefore, the response quantity for academic questions on other types of social media should be studied and compared to provide more accurate and comprehensive research findings on this issue. This research could also be extended to social media in other languages.

Finally, this study only focused on the responses from scholars on academic social media. There are other types of interactions on social media, such as reading, following and recommending, that could also be used as future research objects in multi-dimensional research on the involvement of scholars in academic social media and that could provide theoretical and practical recommendations for improving scholars' participation.

6. Conclusion

To improve the response quantity for questions posed by scholars on academic Q&A platforms, this study uses questions from five disciplines on ResearchGate Q&A as the research object and analyzes which linguistic characteristics affect response quantity. It further differentiates the characteristics affecting the response quantity for questions from different disciplines. The linguistic characteristics with a significant impact on response quantity included positive emotion, second-person pronouns, word count and function words, further confirming that some linguistic characteristics affecting question response

quantities are different in different disciplines. We also discussed the inherent reasons for these effects and pointed out the significance of this research for academic social Q&A platforms and scholars.

This paper only addresses academic questions from a limited set of disciplines on academic social Q&A platforms and does not consider academic questions on other social media. Future research should extend the focus to include other forms of social media where academic questions are posed and use additional methods to conduct more comprehensive and in-depth research on this issue.

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